Directional Consistency in Fine-Grained NLI: Baselines for DiCo-NLI (Track 1)

CS-878 Deep Learning, Assignment 1

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1 Introduction and Task Definition

NLI models can score well by exploiting superficial cues (Gururangan et al., 2018), and standard evaluation scores each ordered pair on its own, so it cannot show whether a model stays coherent when a pair is reversed. DiCo-NLI (SemEval-2027 Task 2) measures both (DiCo-NLI Organizers, 2026). We work on Track 1 (English).

Black-box view. The input is an ordered pair of short phrases (t_1, t_2) and the output is one of four labels: `EQUIVALENCE` (EQ), `FORWARD_ENTAILMENT` (FWD), `BACKWARD_ENTAILMENT` (BWD) or `NEGATIVE_OTHER` (NEG). EQ means each phrase entails the other, FWD that only $t_1 \Rightarrow t_2$, BWD that only $t_2 \Rightarrow t_1$, and NEG collects every other relation. For example, "Japanese PM Shinzo Abe" $\rightarrow$ "Abe" is FWD, and the reversed pair is BWD. A coherent classifier should therefore satisfy $f(t_2, t_1) = \mathrm{rev}(f(t_1, t_2))$, with $\mathrm{rev}(\text{FWD}) = \text{BWD}$, $\mathrm{rev}(\text{BWD}) = \text{FWD}$ and $\mathrm{rev}(\text{EQ}) = \text{EQ}$. NEG rows appear in one orientation only.

White-box view. FWD and BWD differ only in direction, so symmetric signals such as word overlap or cosine similarity cannot separate them. Similarity is also not equivalence: "A sheep" and "A large sheep" share almost everything but the relation is one-way. NEG merges several relations, and the phrases are so short that one modifier often decides the label. There is also an obvious shortcut risk: the more specific phrase is usually longer, so a model could follow length instead of meaning.

Research question and hypotheses. How far do shallow cues and standard fine-tuned encoders get on DiCo-NLI, and how much does consistency reflect correct classification? **H1:** surface cues such as length give non-trivial accuracy and high SoftCons. **H2:** fine-tuned cross-encoders beat the classical baselines on W-F1 and HardCons. **H3:** their remaining errors sit at the NEG boundary rather than on direction.

2 Related Work

Gururangan et al. (2018) showed that hypothesis-only classifiers reach about 67% on SNLI and 53% on MultiNLI, well above chance; our length-only and overlap-only probes apply the same idea to phrase pairs. DiCo-NLI is built from PhrasIS (Lopez-Gazpio et al., 2024), over 10,000 expert-annotated phrase pairs from captions and headlines, and keeps only the reversible relations, mapping the rest to NEG (DiCo-NLI Organizers, 2026). Language models often fail to

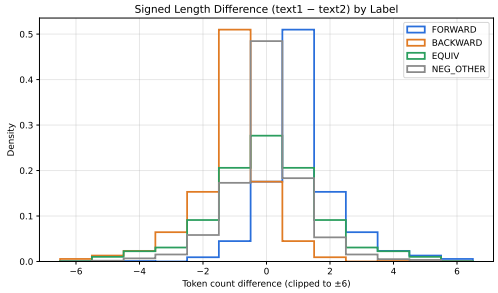


Figure 1: Signed token-length difference $L(t_1) - L(t_2)$ per label.

retrieve a fact backwards (the Reversal Curse; Berglund et al., 2024), though BERT was unaffected in the setting of Wu et al. (2024); DiCo-NLI asks the narrower question of whether a predicted relation transforms correctly under reversal. Its direct predecessor is Apaolaza Larraya et al. (2026), who defined soft and hard coherence (the basis of SoftCons and HardCons) on PhrasIS, found encoders strongest among 17 models, linked errors to lexical overlap and prepositions, and warned that a model can be consistently wrong.

3 Dataset

We use the official split: 3,042 training rows (1,760 `pair_id`s) and 660 development rows (383 pairs). No `pair_id` appears in both splits, so a reversed training pair cannot leak into development. Seven development rows still have an exact text match in training, which we treat as a limitation. Labels are similar in both splits: FWD and BWD about 29% each, EQ 26% and NEG 16%. Because NEG is the smallest class, we also report Macro F1.

4 Exploratory Data Analysis

The full plots are in the separate EDA file; below are the findings that shaped the models.

Length. In FWD rows t_1 averages 3.58 tokens against 2.44 for t_2, and 77% put the longer phrase first; BWD mirrors this and EQ/NEG centre on zero (Figure 1). Length is a real directional signal, but a surface one, so we test it alone as a shortcut.

Overlap and containment. In 45% of entailment rows one phrase's words are fully contained in the other. Mean Jaccard is 0.41 for entailment but only 0.24 for EQ, because paraphrases use different words. Jaccard is symmetric, so we separate it from directional containment features.

POS. Words found only in t_1 of FWD pairs are often adjectives (0.39 per pair vs. 0.16 for EQ) and proper nouns, i.e. the details that make a phrase more specific. We add POS counts of non-shared words.

Similarity and vocabulary. MiniLM cosine is about 0.74 for entailment, 0.72 for EQ and 0.62 for NEG: it separates related from unrelated pairs but is direction-blind, so the encoders must read the pair jointly. 47% of training word types occur once and 15.4% of development tokens are unseen, which favours pretrained subword models over TF–IDF.

5 Feature Engineering

From the EDA we build 26 features per ordered pair: length (token counts and signed token/character differences), symmetric overlap (Jaccard, identical text), directional coverage and containment in both directions, negation and digit mismatch, and spaCy POS counts (noun, proper noun, adjective, verb, preposition, number) of non-shared words on each side. All of

Model	W-F1	SoftCons	HardCons	Macro-F1	Acc.
Majority class	0.129	0.000	0.000	0.112	0.288
LR (length only)	0.464	**0.989**	0.628	0.408	0.532
LR (symmetric overlap only)	0.324	0.368	0.188	0.291	0.370
LR (engineered features)	0.593	0.877	0.653	0.558	0.602
TF–IDF + LR	0.484	0.877	0.542	0.452	0.500
TF–IDF + features + LR	0.620	0.856	0.661	0.588	0.626
BERT-base (3 seeds)	0.705 ± 0.023	0.774 ± 0.016	0.693 ± 0.021	0.680 ± 0.024	0.707 ± 0.020
RoBERTa-base (3 seeds)	**0.765** ± 0.002	0.804 ± 0.011	**0.739** ± 0.006	**0.742** ± 0.002	**0.762** ± 0.004

Table 1: Development-set results with the official scorer. W-F1, SoftCons and HardCons are the official metrics.

them except Jaccard and the identical-text flag change sign or swap roles under reversal. Features are z-scored on the training set.

6 Baseline Models

Baseline A (classical). Logistic regression (scikit-learn, $C = 1$) on: length features only; symmetric overlap only, a probe that has no notion of direction; all 26 features; TF–IDF unigrams and bigrams computed separately for each phrase and concatenated so that order is kept; and TF–IDF plus features. A majority-class model gives the floor.

Baseline B (fine-tuned encoders). `bert-base-uncased` and `roberta-base` are fine-tuned end to end as cross-encoders that read t_1 and t_2 together, BERT with segment IDs. We use AdamW (lr 3×10^{-5}, weight decay 0.01), batch 16, linear warm-up, six epochs, and keep the final epoch by a rule fixed in advance. Each model is run with three seeds (13, 42, 87) on an Apple M4 GPU. An early BERT run dropped `token_type_ids`; after the fix the setup was frozen.

7 Evaluation Setup

We use the organisers' scorer on the 660 development rows, of which 554 form $M = 277$ reversible pairs. Weighted F1 averages per-class F1 by class support; Macro F1 weights classes equally. For pair i with predictions $\hat{y}_{i,1}, \hat{y}_{i,2}$ and gold labels $y_{i,1}, y_{i,2}$:

$$\text{SoftCons} = \frac{1}{M} \sum_{i=1}^{M} \mathbb{1}\big[\hat{y}_{i,1} \neq \textsc{Neg} \wedge \hat{y}_{i,2} = \text{rev}(\hat{y}_{i,1})\big],$$

$$\text{HardCons} = \frac{1}{M} \sum_{i=1}^{M} \mathbb{1}\big[\hat{y}_{i,1} = y_{i,1} \wedge \hat{y}_{i,2} = y_{i,2}\big].$$

SoftCons ignores the gold labels, so a model can be consistent and wrong; HardCons requires both directions to be correct, so HardCons $\leq$ SoftCons.

8 Results

Table 1 gives the main picture. With just four length features, LR reaches 0.464 W-F1 but an almost perfect 0.989 SoftCons. The symmetric-overlap probe, blind to direction, manages only 0.368 SoftCons. Directional features lift W-F1 to 0.593, and adding TF–IDF gives the best classical model (0.620). Both encoders are clearly better, and RoBERTa (0.765 ± 0.002 W-F1) is also far more stable across seeds than BERT. The biggest per-class gain is on Eq (F1 0.515 $\rightarrow$ 0.725 for RoBERTa), which fits the EDA: paraphrases share few words, so they need the

Outcome	TF–IDF + feats	BERT	RoBERTa
Both directions correct	183	186	**204**
Consistent but wrong	**54**	28	16
Same entailment label both ways	4	4	0
NEG in one direction only	35	47	42
Other inconsistency	1	12	15

Table 2: Outcomes for the 277 reversible development pairs (encoders: seed 42).

lexical knowledge that pretraining brings. NEG stays the weakest class for every model. Note also that the encoders have *lower* SoftCons than the classical feature models (0.774–0.804 vs. 0.856–0.877) but *higher* HardCons.

9 Error Analysis

Table 2 splits the 277 reversible pairs by joint outcome (seed 42 encoders vs. the best classical model).

Much of the classical model's consistency is consistent wrongness (54 pairs). RoBERTa has only 16 such pairs and confuses FWD with BWD just twice each way (7 and 9 times for the classical model). Its main problem is predicting NEG for one orientation only (42 pairs), which is exactly why its SoftCons is lower. "same-sex marriage" / "gay marriage" is EQ, but the length-only model, the best classical model and RoBERTa all predict FWD/BWD: consistent, and wrong. "a monochrome photograph" / "a black and white photo" splits the models: TF–IDF plus features gives BWD/FWD, BERT gives EQ/NEG, and only RoBERTa gets EQ both ways. And in RoBERTa's largest error group, related pairs rejected as NEG (69 rows), we find near-paraphrases such as "martial arts clubs" / "Martial Arts Schools". As in Apaolaza Larraya et al. (2026), prepositions also cause errors. Finally, BERT without segment IDs gave the same label to both orientations in 119 of 190 entailment pairs, and in none once they were restored.

10 Discussion

All three hypotheses are supported. Length alone produced near-perfect consistency with mediocre classification (H1); RoBERTa clearly beat the classical baselines on W-F1 and HardCons (H2); and the encoders largely solved direction, leaving the NEG boundary as the main failure (H3). The clearest lesson is that SoftCons cannot be read as correctness: the most consistent model in the study is the one that only looks at phrase length.

Limitations. We evaluate on the public development split (seven rows overlap with training) and did no hyperparameter search. MPS training is not fully deterministic (up to 0.03 W-F1 between identical runs), hence three seeds. Error categories come from heuristic rules, some gold labels are arguable ("Australian senator" / "Australian lawmaker" is NEG), and DeBERTa-v3 was dropped after NaN losses on MPS.

11 Next Steps

The one-sided NEG errors call for predicting both orientations jointly, or for training with a loss that rewards $f(t_2, t_1) = \text{rev}(f(t_1, t_2))$, in the spirit of reverse training (Golovneva et al., 2024). We also plan class weighting for NEG, a DeBERTa-v3 run on CUDA, and a blind test submission.

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